TestBoard_ADC124 - selector lookahead / LPSPI

14 October 2026 | Both sensor connectors disconnected | Charger connected | TPS7A2033PDBVR 3.3 V LDO

Fresh measurements and matched previous-firmware comparison

Current captured revision adc124-selector-lookahead-v1, selector policy lookahead, zero batch-prime frames. A four-channel external-address group now uses four conversions; address 6 uses one IN4 conversion. A complete sweep still contains 50 useful sensor samples across both arrays. Optional VMID batches have four discarded conversions per ADC. The saved previous revision adc124-vmid-settle-v1 used a discarded selector prime in each batch.

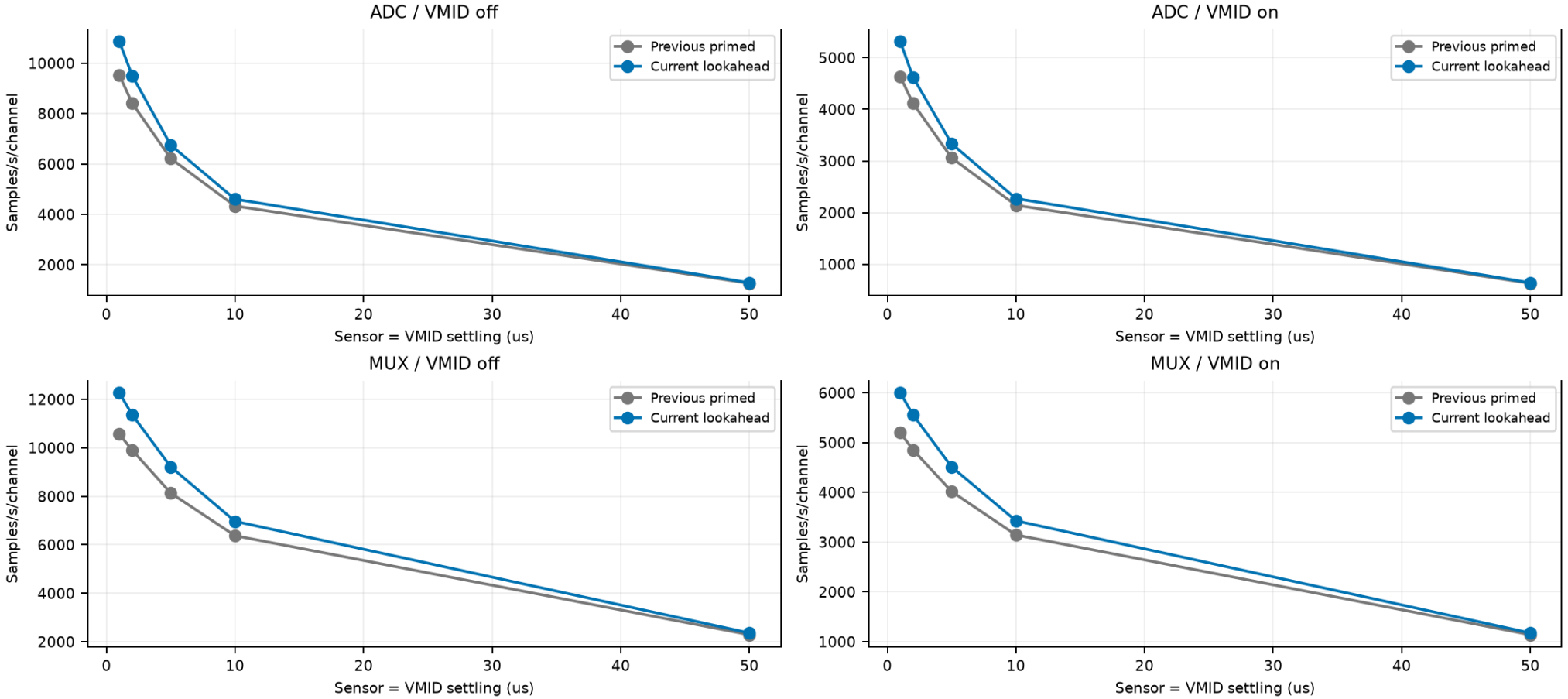
LPSPI only, manual timed acquisition, repeat 1, burst framing. Both ADC and shared-MUX scan modes; sensor settling 1/2/5/10/50 us, VMID off/on, VMID settling equal to sensor settling. Extra tests: sensor 5 us, VMID on, VMID waits 0/1 us; sensor=VMID=5 us at requested 8 MHz versus 16 MHz, VMID off/on. 28 settings, three repetitions, each 1 s warm-up + 15 s measurement. Production firmware only; no DMA/blocking reruns or diagnostic acquisition.

Current captures	Matched previous captures	Current received sweeps	Accounted USB discards	Qualified current quiet records
84	84	7,090,915	415	84

Noise is reported twice: unfiltered measurements include every excursion; quiet-period measurements use the same common all-channel mask as the prior report. Baseline line graphs use full-window medians. Nominal 1 count = 0.805664 mV. The data cannot identify electrical interference sources or prove physical channel identity without known-input stimulation.

5 us / 16 MHz	Previous rate Hz/channel	Current rate Hz/channel	Gain %	Current raw worst SD counts	Current quiet worst SD counts
ADC / VMID off	6,217.0	6,748.7	8.55	0.558	0.558
ADC / VMID on	3,059.4	3,332.8	8.94	0.500	0.500
MUX / VMID off	8,132.6	9,213.3	13.29	0.496	0.496
MUX / VMID on	4,016.1	4,510.5	12.31	0.500	0.500

Sampling rates across settling delays



Requested 16 MHz. One useful sample per physical channel per complete sweep; aggregate useful values/s is 50 times this rate. Rates use delivered device timestamp intervals after warm-up. Accounted USB discards can reduce delivered rate; acquisition counters provide a separate whole-run estimate. Repetition medians are plotted.

ADC scan - matched settling, rates and baseline spread

VMID	Delay us	Old Hz	New Hz	Gain %	Raw worst SD counts	Quiet worst SD counts	Raw worst p-p counts	A spread counts	B spread counts	Worst raw SD route
off	1	9,534.6	10,868.2	13.99	1.467	1.465	28.0	85.0	220.0	2:2:4
off	2	8,412.4	9,510.7	13.06	0.499	0.499	2.0	84.0	198.0	1:1:1
off	5	6,217.0	6,748.7	8.55	0.558	0.558	10.0	81.0	142.0	2:3:0
off	10	4,332.0	4,604.0	6.28	0.500	0.500	1.0	76.0	93.0	2:4:2
off	50	1,264.5	1,286.7	1.76	0.499	0.499	1.0	42.0	13.0	2:4:1
on	1	4,636.4	5,317.7	14.70	0.499	0.499	2.0	33.0	86.0	2:3:2
on	2	4,115.2	4,622.5	12.33	0.499	0.499	2.0	34.0	84.0	2:3:3
on	5	3,059.4	3,332.8	8.94	0.500	0.500	2.0	34.0	76.0	1:3:2
on	10	2,142.0	2,272.3	6.09	0.496	0.496	2.0	35.0	65.0	2:4:0
on	50	630.2	640.9	1.71	0.500	0.500	2.0	28.0	13.0	1:2:3

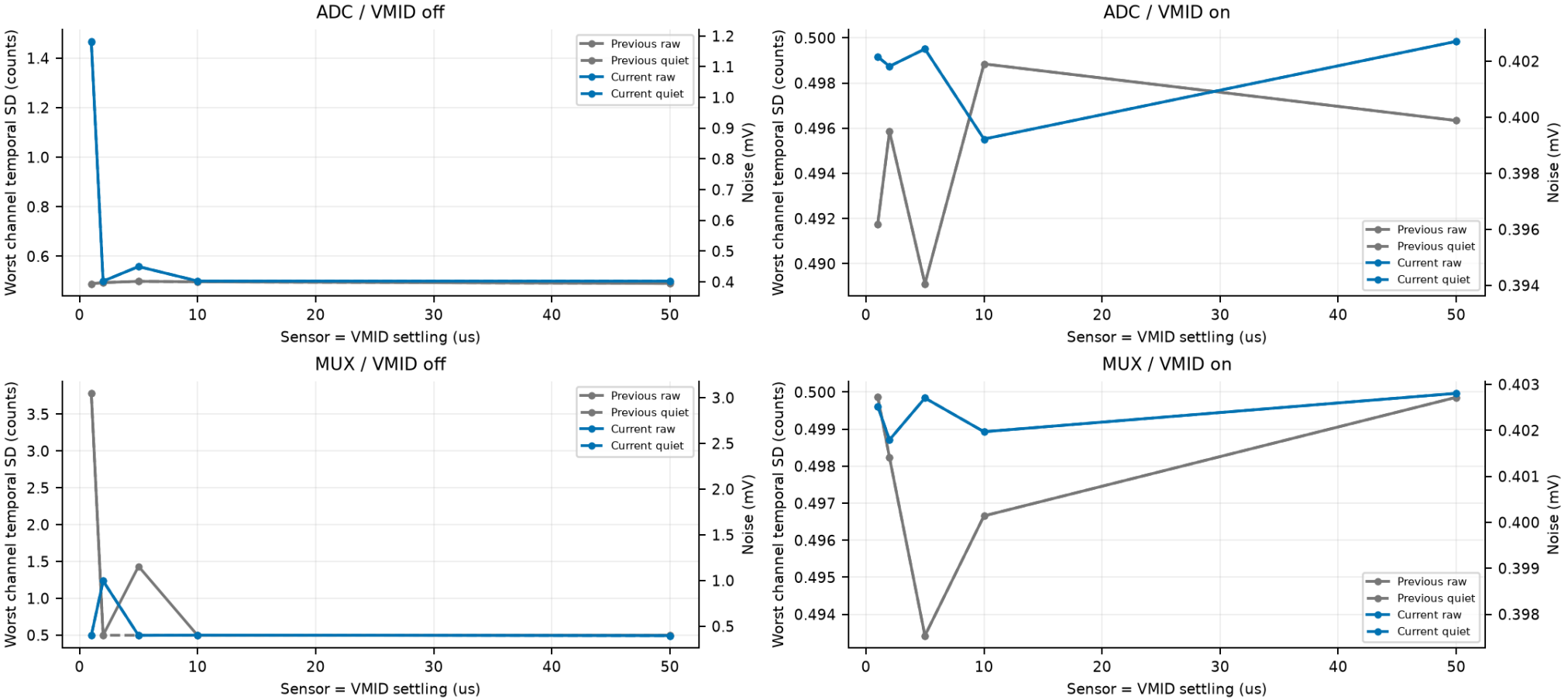
Full-window baseline spread is max-min across the 25 route medians in that array. Temporal SD and baseline spread measure different things. Worst SD/p-p values are the maximum across per-route medians of repetition metrics. Full per-channel data include volts, repetition ranges, baseline population SD and quiet qualification. Neither smoothing nor individual-channel clipping is applied.

MUX scan - matched settling, rates and baseline spread

VMID	Delay us	Old Hz	New Hz	Gain %	Raw worst SD counts	Quiet worst SD counts	Raw worst p-p counts	A spread counts	B spread counts	Worst raw SD route
off	1	10,561.5	12,268.7	16.16	0.495	0.495	7.0	80.0	117.0	2:1:5
off	2	9,904.7	11,362.7	14.72	1.235	1.235	15.0	80.0	109.0	2:3:5
off	5	8,132.6	9,213.3	13.29	0.496	0.496	2.0	77.0	90.0	2:3:4
off	10	6,374.0	6,967.3	9.31	0.500	0.500	2.0	72.0	66.0	1:2:3
off	50	2,288.9	2,361.2	3.16	0.493	0.493	2.0	41.0	9.0	1:1:3
on	1	5,198.4	6,000.0	15.42	0.500	0.500	2.0	33.0	72.0	1:3:2
on	2	4,852.4	5,555.6	14.49	0.499	0.499	2.0	34.0	70.0	2:1:4
on	5	4,016.1	4,510.5	12.31	0.500	0.500	2.0	34.0	64.0	1:4:0
on	10	3,144.3	3,428.9	9.05	0.499	0.499	2.0	34.0	53.0	1:4:6
on	50	1,138.9	1,174.2	3.10	0.500	0.500	2.0	28.0	9.0	1:2:0

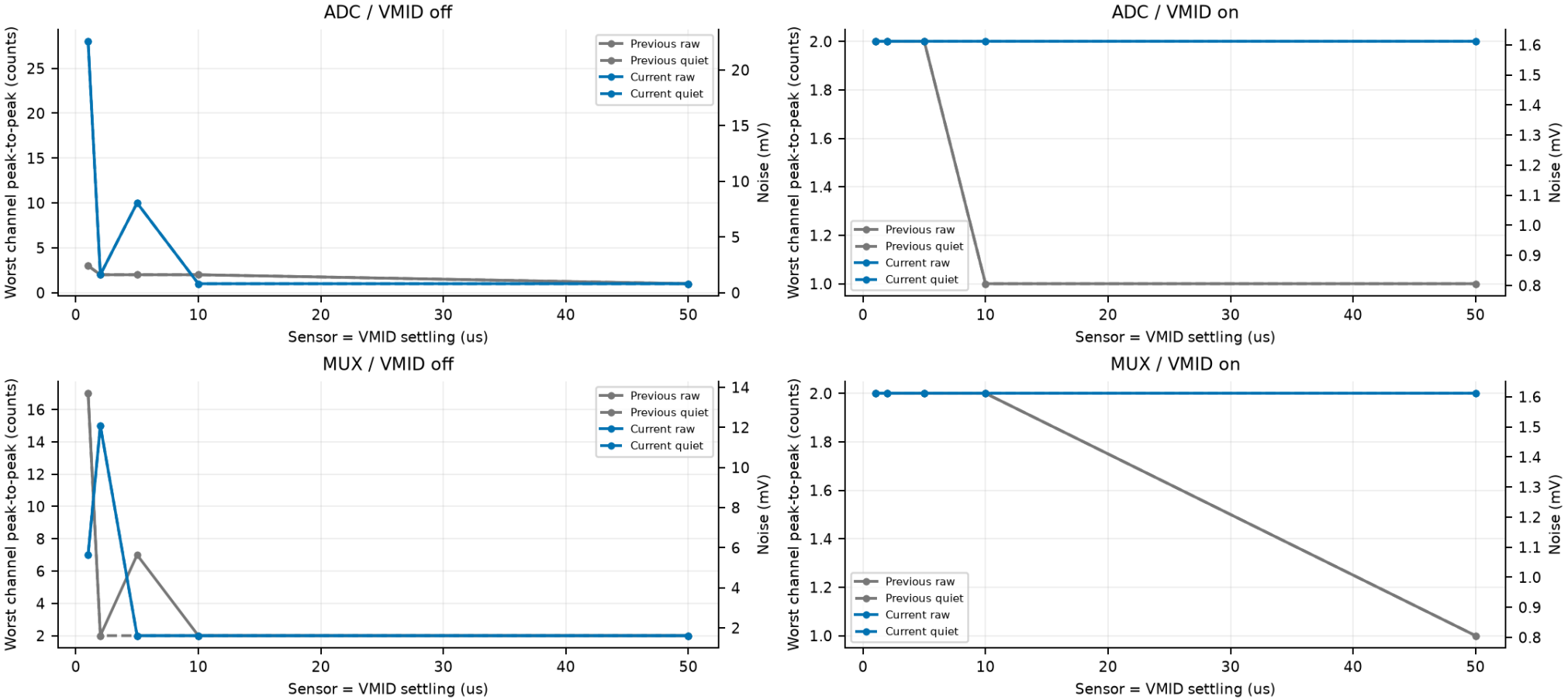
Full-window baseline spread is max-min across the 25 route medians in that array. Temporal SD and baseline spread measure different things. Worst SD/p-p values are the maximum across per-route medians of repetition metrics. Full per-channel data include volts, repetition ranges, baseline population SD and quiet qualification. Neither smoothing nor individual-channel clipping is applied.

Worst channel temporal SD - raw and quiet noise



Solid: unfiltered full windows. Dashed: common quiet windows. Quiet-only metrics do not imply absence of pulses. A zero digital SD means identical recorded codes, not zero analog noise. Peak-to-peak depends on sampled extremes and observation length; both generations have 15 s measurement windows, but quiet retention and sample rates differ.

Worst channel peak-to-peak - raw and quiet noise



Solid: unfiltered full windows. Dashed: common quiet windows. Quiet-only metrics do not imply absence of pulses. A zero digital SD means identical recorded codes, not zero analog noise. Peak-to-peak depends on sampled extremes and observation length; both generations have 15 s measurement windows, but quiet retention and sample rates differ.

ADC - every setting / previous versus current unfiltered noise

VMID	Sensor / VMID us	MHz	Old SD counts	New SD counts	Old p-p counts	New p-p counts	Rate gain %
off	1/1	16	0.489	1.467	3.0	28.0	13.99
on	1/1	16	0.492	0.499	2.0	2.0	14.70
off	2/2	16	0.494	0.499	2.0	2.0	13.06
on	2/2	16	0.496	0.499	2.0	2.0	12.33
off	5/5	16	0.498	0.558	2.0	10.0	8.55
on	5/5	16	0.489	0.500	2.0	2.0	8.94
off	10/10	16	0.497	0.500	2.0	1.0	6.28
on	10/10	16	0.499	0.496	1.0	2.0	6.09
off	50/50	16	0.490	0.499	1.0	1.0	1.76
on	50/50	16	0.496	0.500	1.0	2.0	1.71
on	5/0	16	0.499	0.497	2.0	2.0	11.20
on	5/1	16	0.491	0.499	2.0	2.0	11.39
off	5/5	8	0.495	0.496	1.0	2.0	14.66
on	5/5	8	0.501	0.493	2.0	2.0	14.08

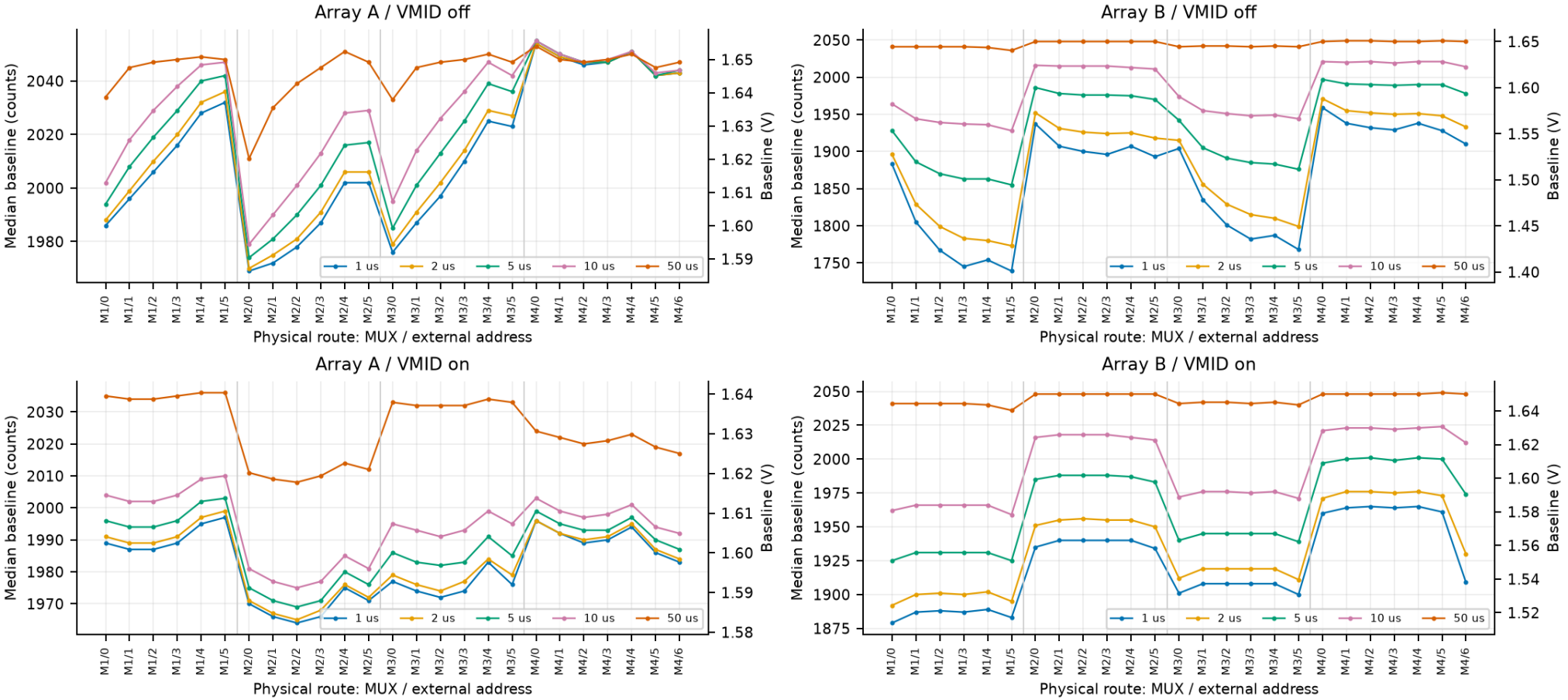
All exact setting matches include explicit VMID delay 0/1 us and requested 8 MHz; neither clocks nor effective VMID delays are pooled. Environmental drift and shorter analog conversion history remain possible contributors to changes.

MUX - every setting / previous versus current unfiltered noise

VMID	Sensor / VMID us	MHz	Old SD counts	New SD counts	Old p-p counts	New p-p counts	Rate gain %
off	1/1	16	3.782	0.495	17.0	7.0	16.16
on	1/1	16	0.500	0.500	2.0	2.0	15.42
off	2/2	16	0.498	1.235	2.0	15.0	14.72
on	2/2	16	0.498	0.499	2.0	2.0	14.49
off	5/5	16	1.433	0.496	7.0	2.0	13.29
on	5/5	16	0.493	0.500	2.0	2.0	12.31
off	10/10	16	0.497	0.500	2.0	2.0	9.31
on	10/10	16	0.497	0.499	2.0	2.0	9.05
off	50/50	16	0.497	0.493	2.0	2.0	3.16
on	50/50	16	0.500	0.500	1.0	2.0	3.10
on	5/0	16	0.815	0.498	3.0	2.0	14.27
on	5/1	16	0.500	0.500	2.0	2.0	13.01
off	5/5	8	0.684	0.500	3.0	3.0	17.56
on	5/5	8	0.670	0.498	3.0	2.0	17.24

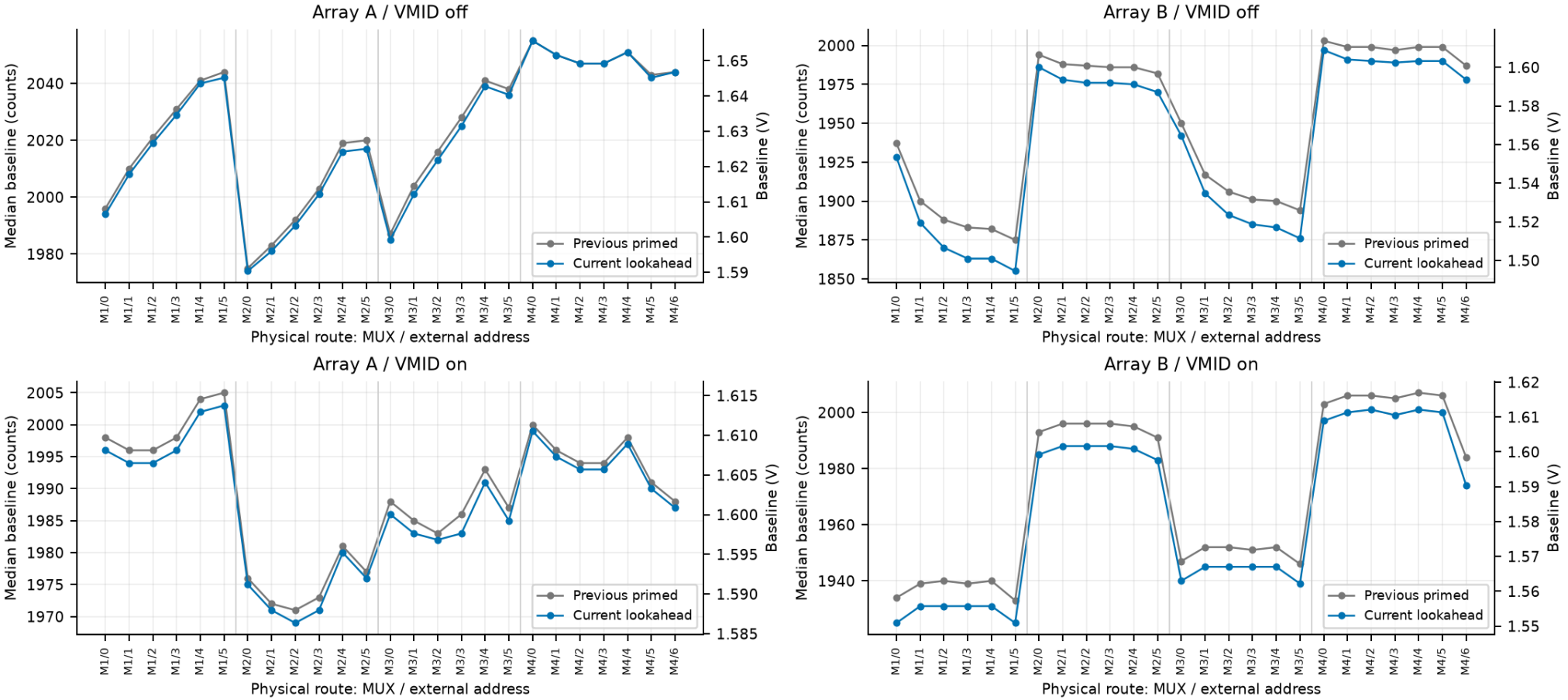
All exact setting matches include explicit VMID delay 0/1 us and requested 8 MHz; neither clocks nor effective VMID delays are pooled. Environmental drift and shorter analog conversion history remain possible contributors to changes.

Every channel baseline - ADC scan / current lookahead



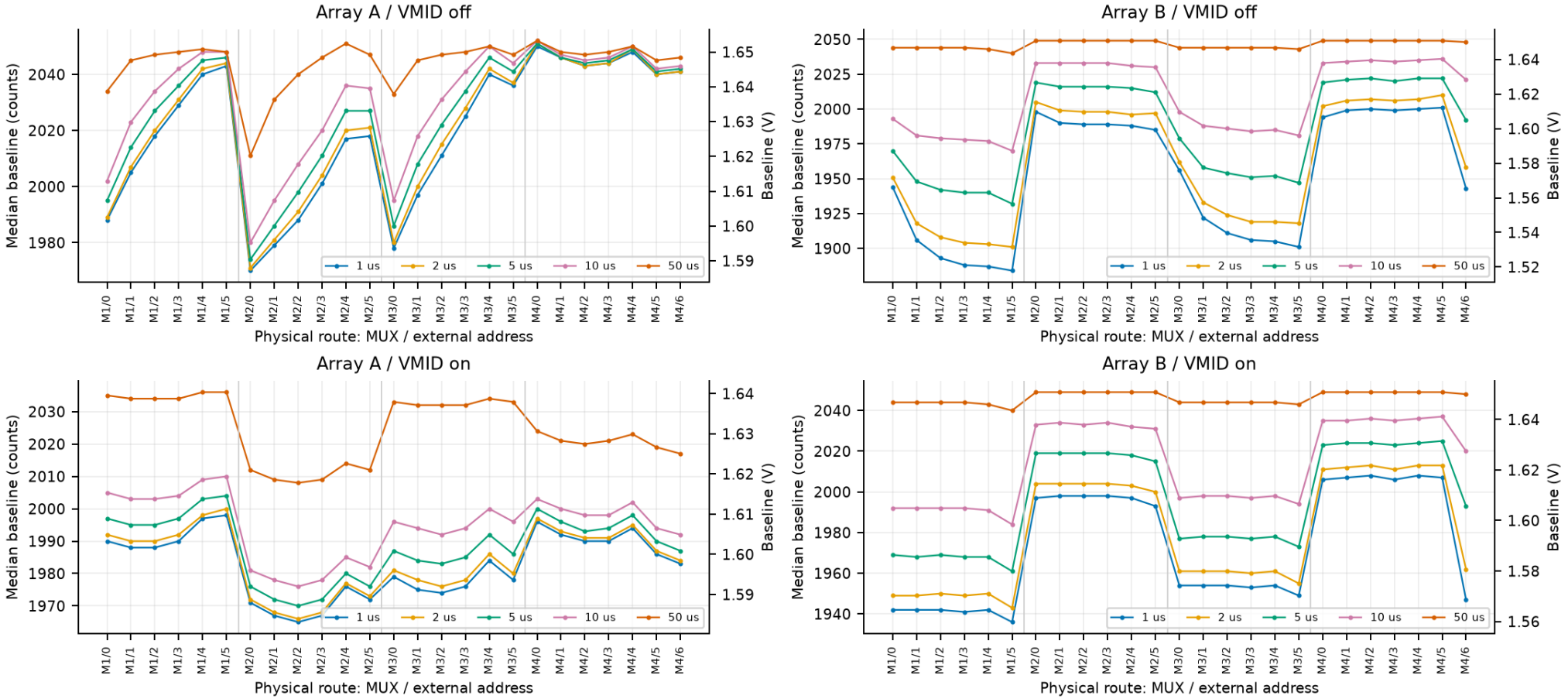
16 MHz, sensor=VMID settling. Full-window per-route median in each repetition, aggregated by median across three repetitions. Lines connect discrete routes for comparison. External addresses are zero-based; MUX1..4 feed ADC IN1..4. Connector channel mappings and bias resistors are retained in the CSV.

Baseline changes - ADC / 5 us / 16 MHz



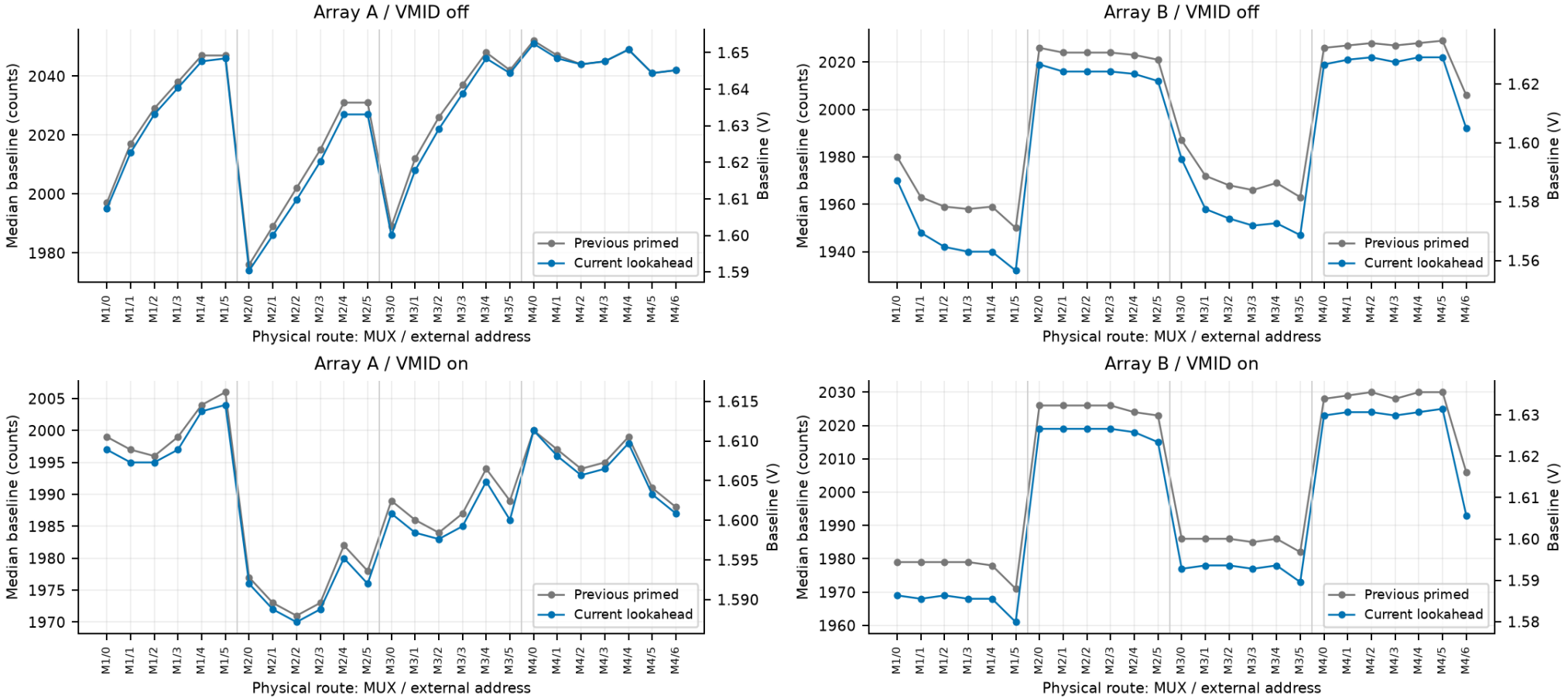
Same physical routes and requested settings. Baseline changes are observations, not calibrated offset measurements. The recordings were sequential and no reference/VMID rail waveforms were measured. All setting-specific route shifts, in counts and volts, are in channel_comparisons.csv.

Every channel baseline - MUX scan / current lookahead



16 MHz, sensor=VMID settling. Full-window per-route median in each repetition, aggregated by median across three repetitions. Lines connect discrete routes for comparison. External addresses are zero-based; MUX1..4 feed ADC IN1..4. Connector channel mappings and bias resistors are retained in the CSV.

Baseline changes - MUX / 5 us / 16 MHz



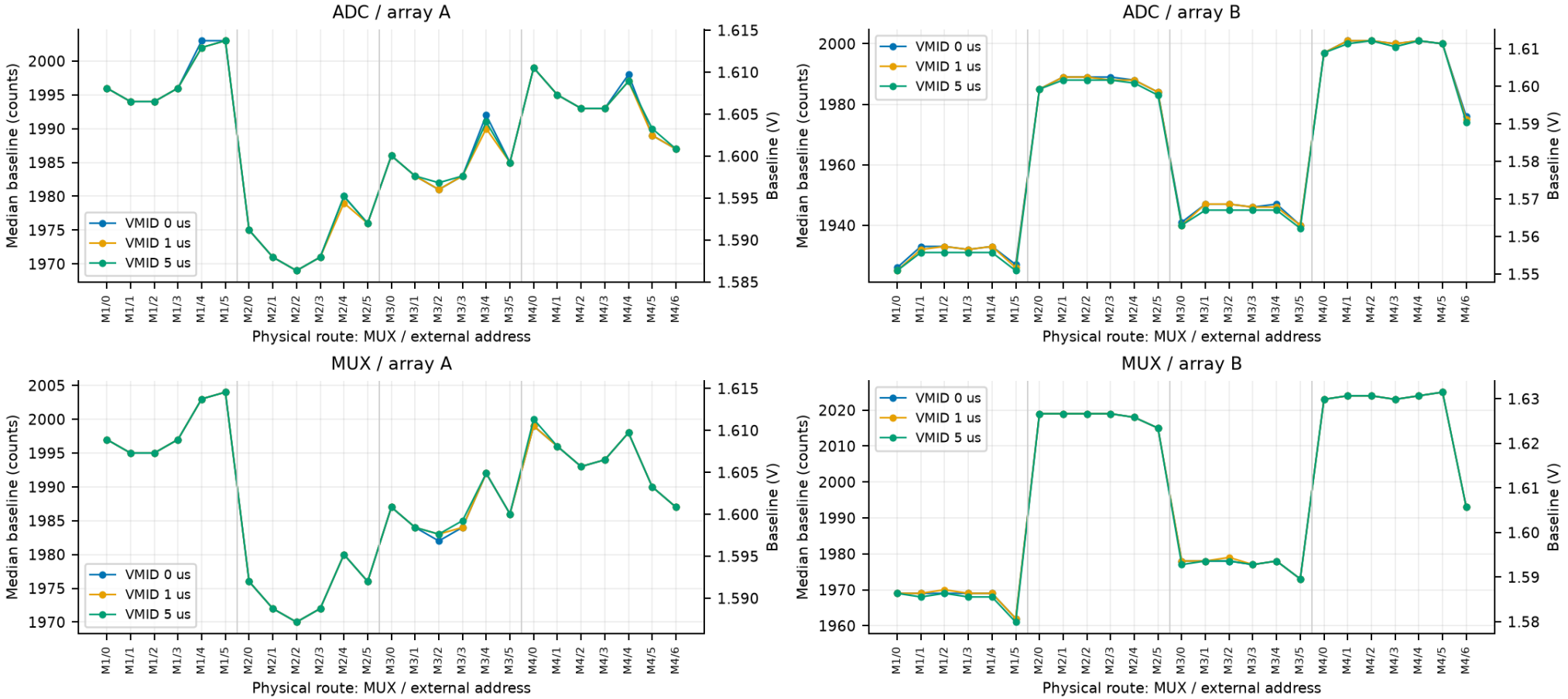
Same physical routes and requested settings. Baseline changes are observations, not calibrated offset measurements. The recordings were sequential and no reference/VMID rail waveforms were measured. All setting-specific route shifts, in counts and volts, are in channel_comparisons.csv.

Independent VMID delay at sensor settling 5 us

Mode	VMID wait us	Previous Hz	Current Hz	Gain %	Raw SD counts / mV	Quiet SD counts / mV	Raw p-p counts / mV
ADC	0	3,899.5	4,336.4	11.20	0.497 / 0.401	0.497 / 0.401	2.0 / 1.611
ADC	1	3,678.1	4,097.0	11.39	0.499 / 0.402	0.499 / 0.402	2.0 / 1.611
ADC	5	3,059.4	3,332.8	8.94	0.500 / 0.402	0.500 / 0.402	2.0 / 1.611
MUX	0	4,694.8	5,364.9	14.27	0.498 / 0.401	0.498 / 0.401	2.0 / 1.611
MUX	1	4,545.5	5,136.9	13.01	0.500 / 0.402	0.500 / 0.402	2.0 / 1.611
MUX	5	4,016.1	4,510.5	12.31	0.500 / 0.403	0.500 / 0.403	2.0 / 1.611

16 MHz, VMID on. Zero omits the VMID analog wait while preserving switching guards and the four discarded VMID conversions. Sensor delay stays 5 us. Settling changes, clock changes and conversion-history changes are separate experimental variables.

Every channel baseline - independent VMID settling



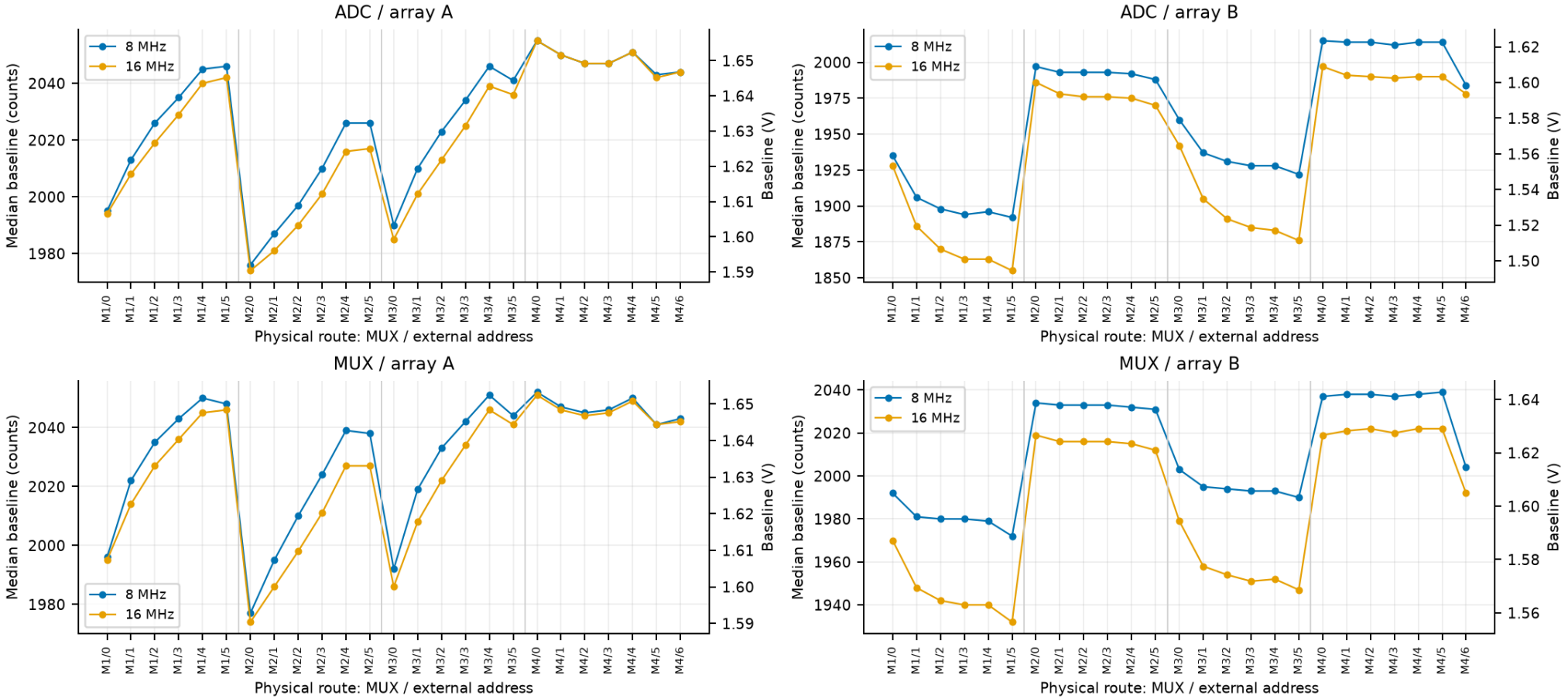
Current lookahead, 16 MHz, VMID on, sensor 5 us. Every point is a full-window median baseline, with all three repetitions contributing.

SPI clock comparison - requested 8 versus 16 MHz

Mode	VMID	Old 8 MHz Hz	New 8 MHz Hz	Old 16 MHz Hz	New 16 MHz Hz	New 8 MHz raw SD	New 16 MHz raw SD
ADC	off	4,401.5	5,046.8	6,217.0	6,748.7	0.496	0.558
ADC	on	2,147.9	2,450.3	3,059.4	3,332.8	0.493	0.500
MUX	off	5,299.7	6,230.5	8,132.6	9,213.3	0.500	0.496
MUX	on	2,579.5	3,024.2	4,016.1	4,510.5	0.498	0.500

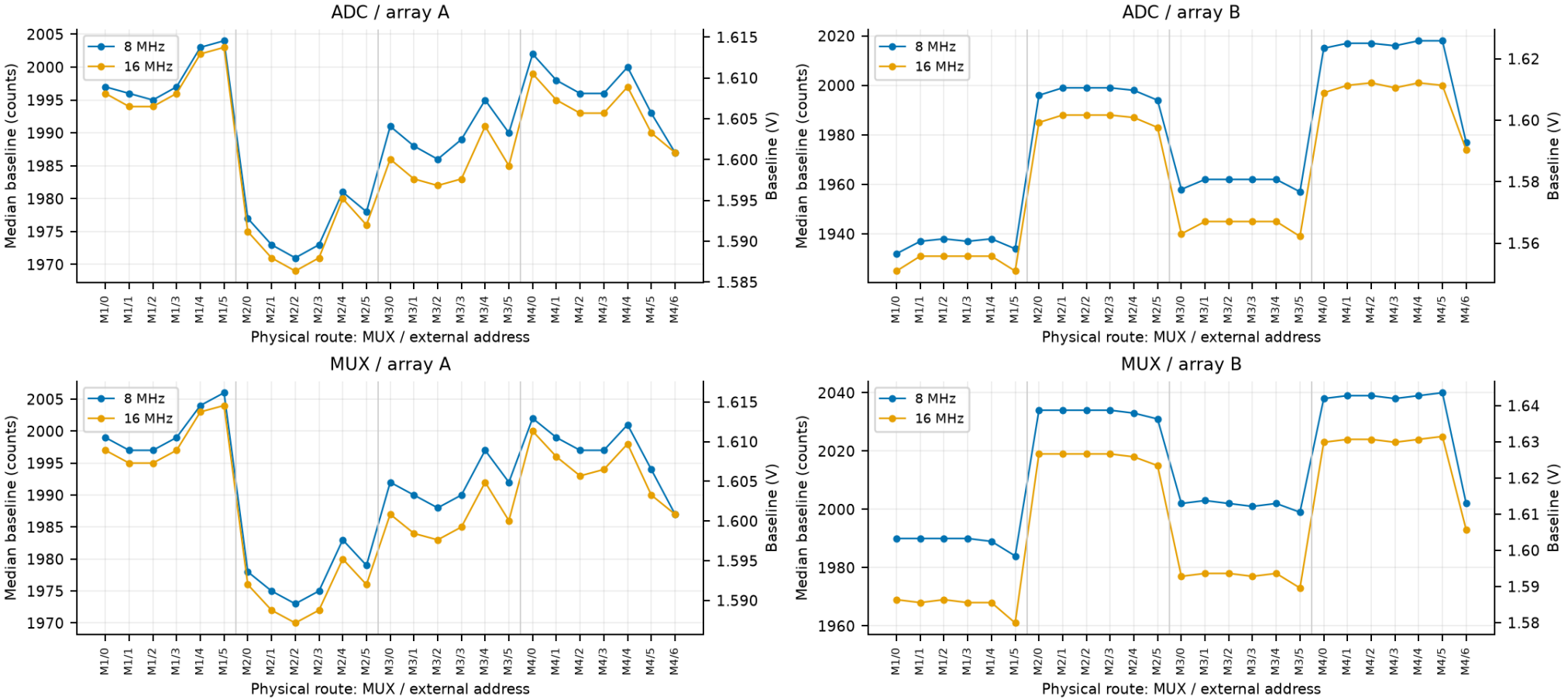
Sensor=VMID settling 5 us. Both VMID off/on. The faster request also changes analog sampling history; it is not a transport-only change. Actual SCLK was not probed.

Every channel baseline - SPI clocks / VMID off



Current lookahead, sensor=VMID settling 5 us. Full-window baseline medians across three repeats.

Every channel baseline - SPI clocks / VMID on



Current lookahead, sensor=VMID settling 5 us. Full-window baseline medians across three repeats.

Conversion budgets and measured settling slopes

Full sweep / repeat 1	Previous words	Current words	Words removed	Sensor / VMID waits: ADC	Sensor / VMID waits: MUX
VMID off	64	50	14	14 / 0	7 / 0
VMID on	134	106	28	14 / 14	7 / 7

Four-input sensor and VMID groups now use four words, instead of five. The sensor group at address 6 uses one word instead of two. Nominal clocking saved: 14 us off / 28 us on at 16 MHz, or 28/56 us at 8 MHz. These are arithmetic budgets, not measured GPIO/SPI edge timing. Both ADC and MUX modes use one shared SPI bus. Startup selector initialization is outside steady-state sweep counts.

ADC mode finishes array A before array B. MUX mode switches both banks at each address, shares one settling wait and reads ADC A then B. Bank B therefore continues settling during A's read batch. Removing the prime changes this additional analog dwell as well as the SPI word count; equal configured delays do not guarantee identical effective dwell between revisions.

Firmware	Mode	VMID	Fit intercept us	Measured us/us slope	Expected slope	Max fit residual us
previous	ADC	off	90.864	13.9996	14	0.018
lookahead	ADC	off	77.694	13.9886	14	0.540
previous	ADC	on	187.152	27.9927	28	0.542
lookahead	ADC	on	160.122	28.0019	28	0.208
previous	MUX	off	87.437	6.9882	7	0.584
lookahead	MUX	off	73.966	6.9893	7	0.553
previous	MUX	on	178.409	13.9923	14	0.630
lookahead	MUX	on	152.071	13.9897	14	0.606

Fit period = intercept + slope x matched sensor/VMID delay. Intercepts contain conversion clocking, GPIO/software work, interrupts and enqueue gaps. No diagnostic profiling was run, so this report does not assign those residuals to individual phases.

Complete selected-route noise audit - MUX/LPSPI/VMID off

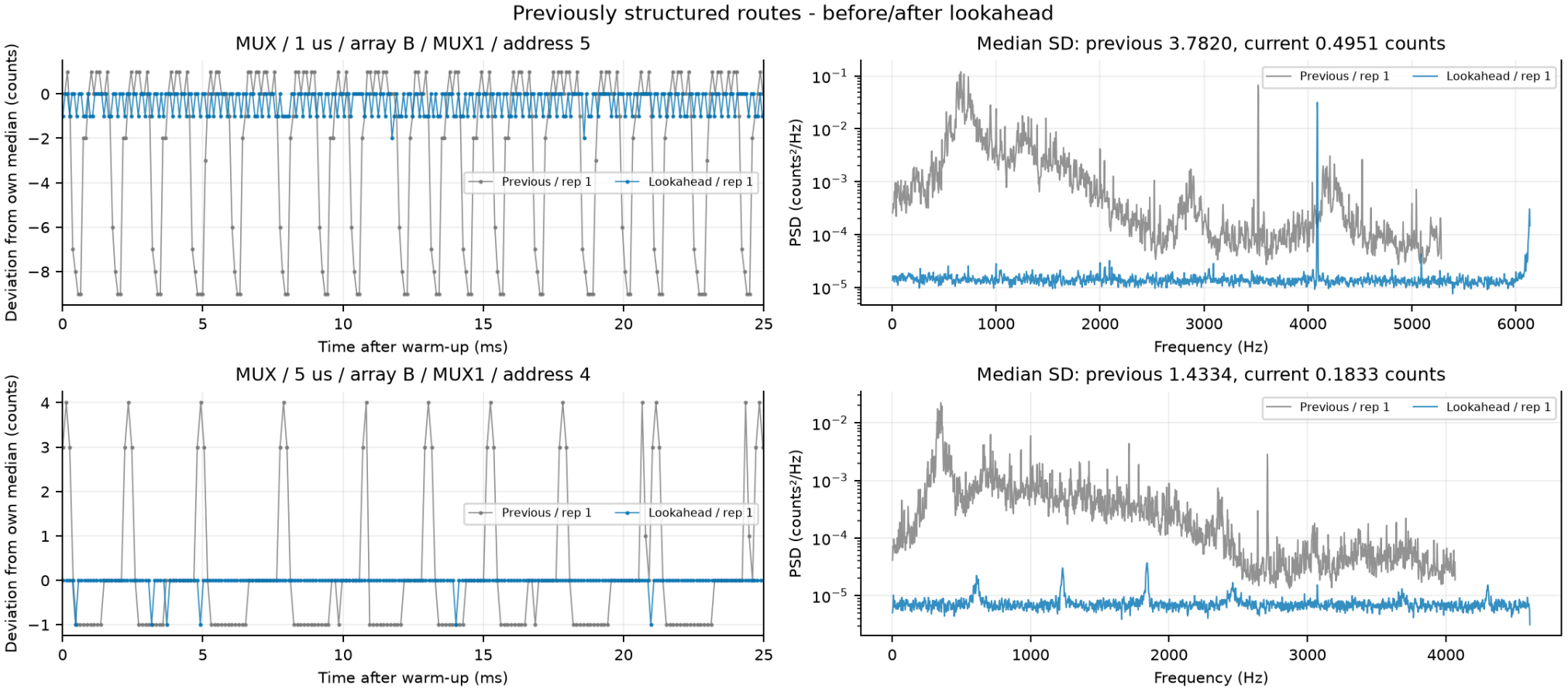
Requested 16 MHz, sensor settling 1/5 us. All 50 routes at each delay, three repeats. Select median unfiltered SD >0.5 counts and max absolute autocorrelation over lags 1-64 >=0.05 in at least two repeats. Listed cases are descriptive evidence of temporal structure, without identifying cause. Below-threshold routes can also be structured.

No current routes meet both the >0.5-count median SD and reproducible-pattern criteria in these two settings.

Qualifying routes by delay: 1 us: 0, 5 us: 0. Route addresses are zero-based; connector sensor/channel labels remain physical positions despite sensors being disconnected. One count = 0.805664 mV. PSD section/window coverage is retained in noise_psd_coverage.csv.

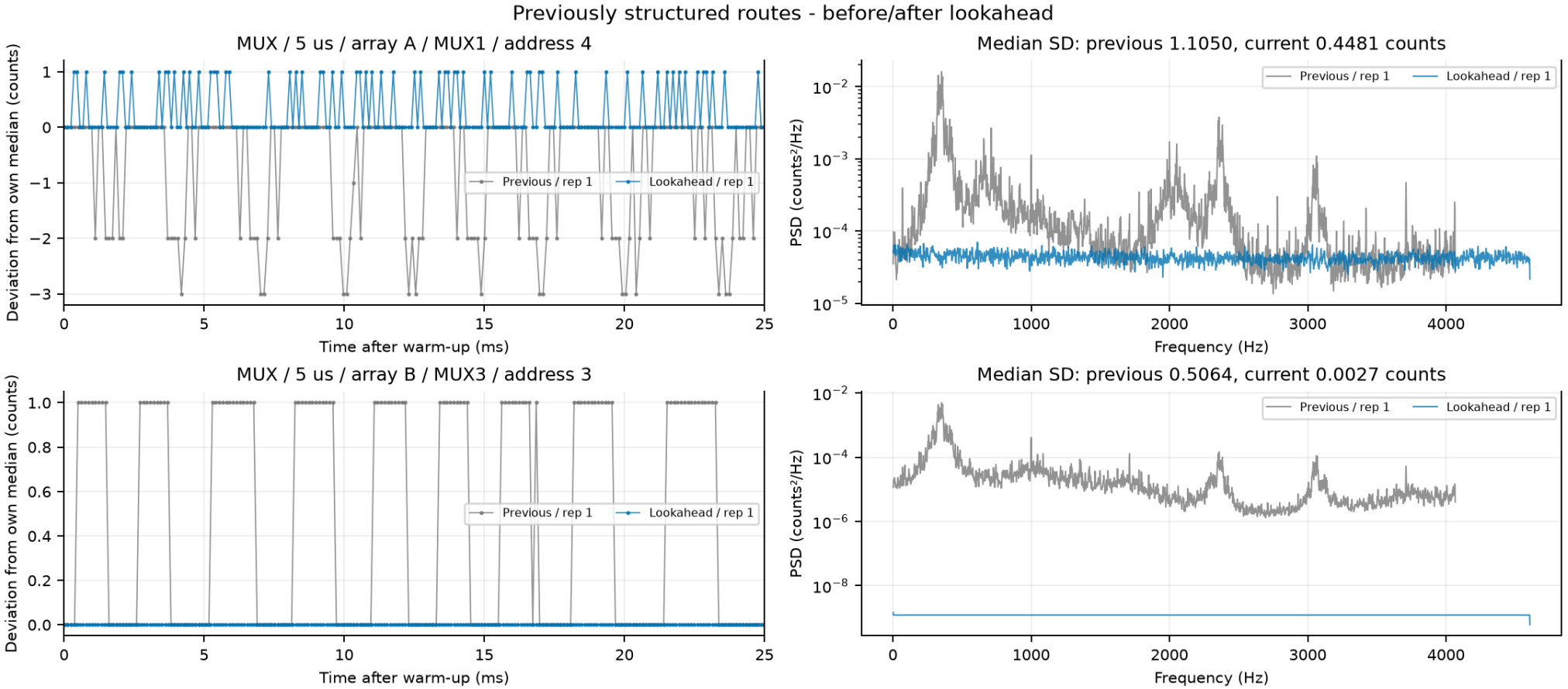
The time/PSD figures on following pages include every selected current route. Original problem routes are also compared separately. SD and PSD use full unfiltered recordings, rather than quiet-only selection.

Previously structured routes - before/after lookahead



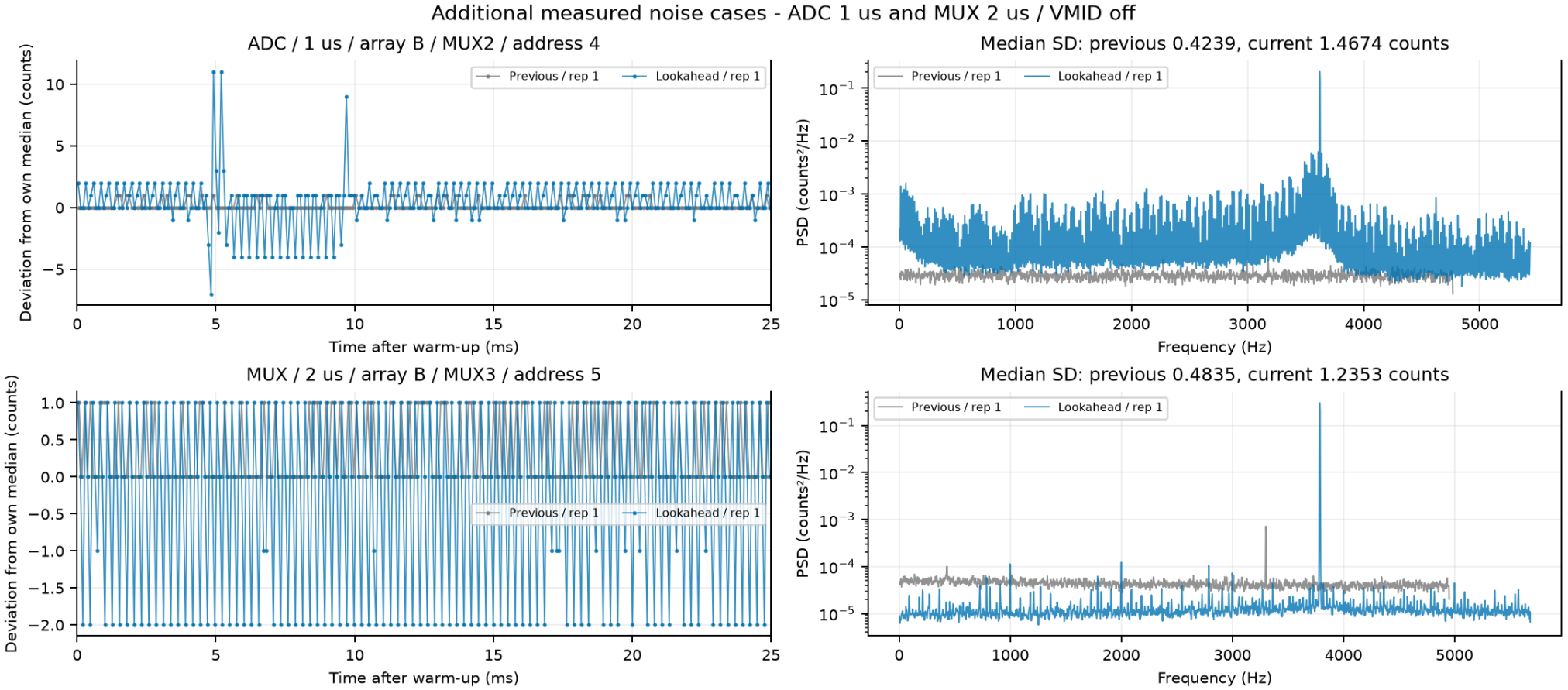
Previous and current first repetitions are shown. Left: unfiltered first 25 ms after warm-up, referenced to each full-record median. Right: Welch PSD from the 15 s capture. 4096-sample Hann windows, 50% overlap, window means removed; split timestamp gaps >1.5 times median period, exclude sections shorter than 4096, unused trailing samples omitted. Normal timestamp jitter remains; frequency axis uses approximate sweep timing. No quiet mask or sample clipping.

Previously structured routes - before/after lookahead



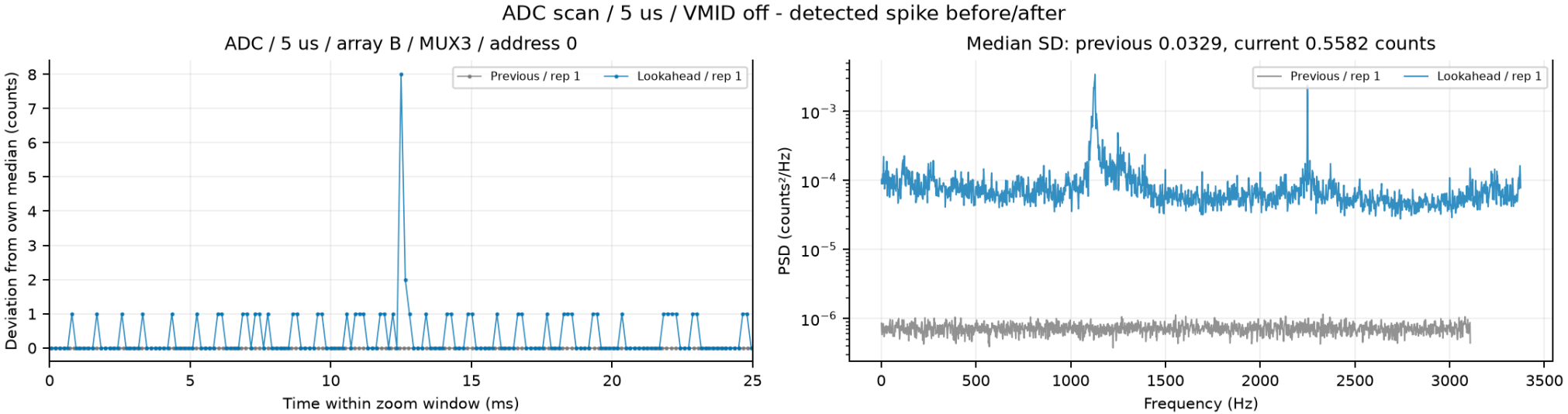
Previous and current first repetitions are shown. Left: unfiltered first 25 ms after warm-up, referenced to each full-record median. Right: Welch PSD from the 15 s capture. 4096-sample Hann windows, 50% overlap, window means removed; split timestamp gaps >1.5 times median period, exclude sections shorter than 4096, unused trailing samples omitted. Normal timestamp jitter remains; frequency axis uses approximate sweep timing. No quiet mask or sample clipping.

Additional measured noise cases - ADC 1 us and MUX 2 us / VMID off



Previous and current first repetitions are shown. Left: unfiltered first 25 ms after warm-up, referenced to each full-record median. Right: Welch PSD from the 15 s capture. 4096-sample Hann windows, 50% overlap, window means removed; split timestamp gaps >1.5 times median period, exclude sections shorter than 4096, unused trailing samples omitted. Normal timestamp jitter remains; frequency axis uses approximate sweep timing. No quiet mask or sample clipping.

ADC scan / 5 us / VMID off - detected spike before/after



Previous and current first repetitions are shown. Left: unfiltered 25 ms window starting 9.759701 s after warm-up, selected around the largest absolute excursion in the current first repetition and using the same elapsed-time interval from the previous capture, referenced to each full-record median. Right: Welch PSD from the 15 s capture. 4096-sample Hann windows, 50% overlap, window means removed; split timestamp gaps >1.5 times median period, exclude sections shorter than 4096, unused trailing samples omitted. Normal timestamp jitter remains; frequency axis uses approximate sweep timing. No quiet mask or sample clipping.

Noise method, verification and final state

Unfiltered SD and p-p use all measured samples after warm-up. Quiet values reuse the prior common-mask method: 50 ms bins, median channel SD in each of eight array/MUX groups; group threshold max(0.6 counts, P20+0.15, lower40% median+4 x 1.4826 x MAD). Reject a bin exceeding any group threshold and its neighbors. Then reject coherent separated secondary-state excursions on at least two routes, plus one neighboring sweep on each side, identically for all 50 channels.

The fine detector anchors dominant one-code states (>=60%) or adjacent two-code states (>=80%), allows one extra count, and requires >=1% tail mass plus a separated histogram peak >=0.5% and >=3 times edge density. Unimodal Gaussian tails and isolated route variation remain. A quiet record requires >=1 s and >=100 retained frames. Retained seconds credit observed intervals only, capped at median period. These masks do not establish a charger cause or guarantee white noise.

Per-setting noise/baseline values are per-route medians of three repetition statistics; maxima are selected independently across routes. Baseline spread is max-min of route baseline medians, while population SD across baselines is also saved. Both differ from temporal noise. Counts use nominal 3.3/4096 V; voltage/reference calibration was not performed.

Generation	Captures	Qualified quiet records	Firmware error totals
previous	84	84	0
lookahead	84	84	0

Each stream passes raw SHA-256, two independent binary decoders, route labels, 12-bit ranges, status/configuration, nonoverlapping timestamps and sent/received accounting. Firmware intentionally discards complete sweeps under USB pressure; received must equal sent, and acquired equals sent+discarded. Both firmware builds passed; 81 focused Python checks and two native firmware checks passed. Native checks include exact lookahead conversion counts and digital payload mapping; physical wiring requires known-input stimulation.

Lifecycle: complete_production_restored_verified. Final production revision adc124-selector-lookahead-v1, running=false. Original settings restored. Frozen source/HEX images and hashes, command/status logs and raw data accompany the dataset. Only the frozen current production image was uploaded for this rerun; no firmware source changes were made.

Data and reproducibility

Current raw: docs\testing\datasets\testboard-adc124\selector_lookahead_lpspi_2026-10-14

Matched reference raw: docs\testing\datasets\testboard-adc124\upgrade_disconnected_2026-10-14\production

File	Contents
analysis.json / analysis_cache.json	Verified structured results and hash-bound per-capture cache
capture_metrics.csv / summary.csv	Quiet retention, transport accounting, timing and setting summaries
channel_summary.csv / channel_repetitions.csv	Quiet route baseline, sample SD and p-p, counts and volts
unfiltered_*	Every-sample capture/setting/channel metrics, including temporal correlations in repetition records
firmware_comparisons.csv	All 28 exact-setting rate and unfiltered-noise comparisons
channel_comparisons.csv	Every route baseline shift and unfiltered noise comparison
delay_fits.csv	Measured matched-delay slopes/intercepts, both revisions
*quiet_mask.npz / *fine_quiet_states.json	Common raw-frame masks and detector parameters
selected_noise_current.json / selected_noise_audit.csv	All-route MUX/LPSPI/VMID-off 1/5 us pattern audit and qualifying cases
figures/	Baseline line graphs, rate/noise comparisons and raw time/PSD plots

Reproduce with benchmark_adc124_lookahead_lpspi.py, analyze_adc124_lookahead_lpspi.py and report_adc124_lookahead_lpspi.py in Arduino_Sketches/TestBoard_ADC124/benchmarks. Preserve both disconnected generations before the later connected-sensor experiment. No commit or push was performed.